# COMSATS University Islamabad, Lahore Campus

**Terminal Lab Examination – Fall 2025**

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| Course Title: | Web Technologies | | | Course Code: | CSC336 | Credit | 3(2,1) |
| Course Instructor/s: | Asif Shahzad | | | Programme Name: | BSCS | | |
| Semester: | 5 | Batch: | FA23 | Section: | G2 | Date: | Dec. 16, 2025 |
| **Time Allowed:** | **1.5 Hours** | | | **Maximum Marks:** | **50** | | |
| Student's Name: | | | | Reg. No. | | | |

**Important Instructions:**

1. Store data in *products.json* file. Or use some online database and store images in */public/images/* folder.

2. I would open only home page using: *localhost:3000/* All other pages shall be properly linked and must not require entering explicit URL in browser. Evaluation would be done in lab with viva.

**3. Once you are done with you work. Make a copy of it. Delete node_modules folder from your project before you upload. Do not remove any other file, otherwise >*npm init* command would not work and I would not be able to setup. This is very important. There must be at least 2 products info already added when you upload your work.** For name of the file that you upload, use format **<your registration number>-<your name>.zip**

Question 1: *CLO 5*

Create an e-commerce store that shall have following features. Feel free to use AI tools.

**Product Management (15 Marks)**
Implement Add, Edit, and Remove Product features. Product fields: title, price, description, primary image, and multiple secondary images. On the Edit page, allow updating or deleting images. Image deletion must use AJAX. Show **Add Product** link in the header and **Edit/Delete** links with each product on the home page (/). Implement Delete using AJAX.

**Product Listing & Cart (15 Marks)**
Display all products on the home page with title, price, and primary image. Clicking the title or image opens a detail page showing all product information, a quantity field, and an **Add to Cart** button. Implement the cart using session. Add a **View Cart** button in the header to show all cart items, per-product total, gross total, and an **X** link to remove items, do using AJAX.

**Admin Authentication (15 Marks)**
Provide an **Admin Login** link in the header. Only logged-in admins can Add, View, or Remove products. Show these links and **Logout** in the header after login. If a non-admin accesses admin pages or handlers, display an appropriate error message. Fix username/password to *admin/123.*

**UI/UX (5 Marks)**
Ensure the design is clean, responsive, and visually appealing.